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Impact of Supervised-concurrent Exercise During Pregnancy on Infant Neuromotor Skills: A Post-hoc Analysis Stratified by Maternal BMI

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Abstract

Background This is a post-hoc secondary analysis of a randomized controlled trial whose purpose was to analyze the effect of a supervised-combined aerobic and resistance exercise (concurrent exercise) training program during pregnancy on 1-month infant neuromotor skills based on maternal pre-pregnancy body mass index (BMI).

Methods Ninety-four pregnant women participated in this study, which was conducted at East Carolina University (Greenville, North Carolina, USA) between 2015 and 2018, and were allocated into concurrent exercise ($n=42$) or stretching and breathing group ($n=52$). The exercise group followed a 50-min 3 days/week moderate intensity concurrent exercise training program from the 16th gestational week until birth. Infant neuromotor skills were assessed with the Peabody Developmental Motor Scales, 2nd edition (PDMS-2).

Results Per protocol analyses showed that infants from the exercise group and whose mothers were normal-weight, had higher PDMS-2 Stationary, Locomotion (borderline) and Gross Motor Quotient (GMQ) percentiles than infants of stretching and breathing participants ($p=0.022$, $\eta p2=0.20$; $p=0.054$, $\eta p2=0.15$; $p=0.022$, $\eta p2=0.20$, respectively). No differences between groups were found in infants whose mothers were normal-weight in reflexes percentiles in the adjusted model, or those infants whose mothers were overweight/obese in any of the outcomes (all $p \geq 0.05$).

Conclusions Concurrent exercise training during pregnancy improves neuromotor skills in infants at 1 month of age in normal-weight women. Further studies are needed to understand the influence of concurrent exercise training during pregnancy on infants whose mothers are overweight or obese, and the potential mechanism behind the role of maternal BMI in the development of infant neuromotor skills.

Clinical Trial Registration ClinicalTrials.gov Identifier: NCT03838146.

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Key Points

Infants from normal-weight (but not overweight/obese) women who performed concurrent exercise during pregnancy presented better neuromotor skills than infants of stretching and breathing participants.

The results of this study encourage the implementation of strategies to keep pregnant women active throughout pregnancy, to improve the neuromotor skills of future generations.

Further studies are needed to understand the influence of concurrent exercise training during pregnancy on infants whose mothers are overweight or obese, which provides future lines of research necessary for society.

Keywords Exercise, Pregnancy, Neurodevelopment, Child

Background

In the absence of contraindications, women should be encouraged to engage in combined aerobic and resistance exercise (concurrent exercise) before, during, and after pregnancy to improve maternal and offspring health outcomes [1]. Although there is little research regarding the influence of exercise during pregnancy on infant health, previous studies have shown associations with a better metabolic intrauterine environment [2], control of birth weight in normal ranges [3], enhanced Apgar percentile [4], and lower risk of preterm birth [5].

Further, several studies have shown that regular maternal exercise induces benefits for the fetal brain development [6–8]. Knowing if exercise during pregnancy can improve neonatal neurodevelopment would be of great importance in public health to prevent future neurodevelopmental diseases. Nevertheless, there is little evidence addressing the topic to date.

Neuroplastic changes in the brain and nervous system start from conception through pregnancy and after birth, particularly through the first 1000 days of life. These times are critical, when neurological pathways influencing brain structure and function are developing [9, 10].

Although evidence is scarce, maternal lifestyle/environment during pregnancy, which includes prenatal exercise and maternal body composition, seems to be a key determinant for offspring neurodevelopment [10]. Animal studies have shown that exercise during pregnancy is correlated with enhanced neurogenesis, brain plasticity, and increased learning and memory cells [11, 12]. In humans, several studies have assessed how exercise during pregnancy could influence children's neurodevelopment [13–15], but only three studies [14, 16, 17] examined neonates (< 1 month of age). These three studies demonstrated that neonates of mothers who exercised during pregnancy had higher levels of orientation and self-regulation [14], neuromotor skills (PDMS-2 Locomotion percentile) [16], and brain maturation [17]. Nevertheless, these previous studies were limited to maternal aerobic exercise and did not distinguish between normal-weight (BMI from 18.5 to 24.9 kg/m2) and overweight/obese mothers (BMI above 25.0 kg/m2). In addition, only McMillan et al. [16] assessed neuromotor skills. The effect of concurrent exercise during pregnancy on infant neuromotor skills, as well as the influence maternal BMI on infant neuromotor skills, is not known. Given the former, the purpose of this study is to conduct a post-hoc analysis to examine the effect of a supervised-concurrent exercise training program from the 16th gestational week until delivery on 1-month infant neuromotor skills based on maternal pre-pregnancy BMI.

Materials and Methods

We conducted a post-hoc secondary analysis of a randomized controlled trial (RCT) to examine the effect of concurrent exercise during pregnancy on infant neuromotor skills. The initial RCT was designed to assess the influence of different exercise modes (aerobic, resistance and concurrent exercise) compared to non-exercise on infant health outcomes [18]. All protocols were approved by the East Carolina University Institutional Review Board in accordance with the Declaration of Helsinki for experiments involving human participants. Women were recruited via study fliers at local obstetric clinics between July 2015 and January 2018. Pregnant women were screened and enrolled between 13- and 16-weeks of gestation. All participants signed an informed consent.

Participants

We focused recruitment on low-risk, healthy women, 18 to 40 years of age, pre-pregnancy BMI above 18.5 kg/m^2, with a singleton pregnancy, and no contraindications regarding the practice of exercise. All women were healthy, non-smokers, with no current alcohol, drugs, or medication use. Criteria for exclusion included smoking, pre-existing diabetes mellitus, hypertension, cardiovascular disease, as well as comorbidities known to affect fetal growth and well-being (i.e., systemic lupus erythematosus). All exercise protocols and study visits were conducted at East Carolina University (Greenville, North Carolina, USA) between 2015 and 2018.

Randomization

Prior to 16 weeks gestation, and after written informed consent and physician clearance for exercise was obtained, participants completed a modified Balke submaximal treadmill test, previously validated and

replicated for pregnant women by Mottola et al. [19]. Results from the treadmill test were used to determine individual aerobic capacity and calculate specific target heart rate ranges for moderate-intensity exercise training, as previously published [19]. Heart rate (POLAR FS2c, Polar Electro Inc., Bethpage, NY) and 12–14 rating of perceived exertion (RPE) [20] was monitored before, during, and after each training session. One repetition maximum test for major muscle groups were also completed to assess maternal strength prior to exercise training.

After completing these tests, participants were randomized into one of four groups (aerobic, resistance exercise, concurrent exercise, non-exercising control) using GraphPad software random sequence generator (GraphPad software, San Diego, CA) that was concealed prior to group assignment. In this study, only data from the concurrent exercise and non-exercising stretching and breathing groups were included. Concurrent exercise during pregnancy has been proved by literature to have more favorable effects in moth maternal and infants' health than aerobic or resistance exercise alone [21, 22]. Study personnel completing infant neuromotor assessments were blinded to participant group assignment; however, due to the nature of an exercise intervention, participants and exercise trainers could not be blinded to group assignment.

Exercise Protocols

All participants were supervised by trained exercise instructors in the university facilities and followed a standard protocol. All sessions started at enrollment and were performed three times a week until delivery. All exercise sessions began with a 5-minute warm-up, 50 min of their randomized group activity, and ended with a 5-minute cool-down. The concurrent exercise training group performed, individually, five circuits lasting 4.5–5 min, each. Based on individual preference, aerobic exercise modalities included treadmill, stationary bicycle, elliptical, or aerobics at moderate intensity (40–59% heart rate reserve, 12–14 RPE). To maintain the appropriate HR zone, speed and grade were adjusted on the treadmill, and resistance and speed levels were adjusted on the elliptical and recumbent bike. Resistance exercises were performed with sets of 15 repetitions of each exercise at a moderate intensity (60% of 1 repetition maximum. adjusted as needed to maintain their progression). Seated machines (leg extension, leg curl, shoulder press, chest press, triceps extension, latissimus dorsi pull down), dumbbells (biceps curls, lateral shoulder raises, front shoulder raises), resistance bands, exercise balls, benches, and/or mats were used. In per protocol analyses (attendance ≥ 75% of the intervention), the average attendance at the concurrent exercise intervention was 89.9%. In intention-to-treat analyses, the average attendance was 82.2%.

Participants in the stretching and breathing group completed a 50-minute supervised stretching and breathing routine, three times per week. Stretching movements targeted major muscle groups and breathing exercises. Low intensity was confirmed during sessions using a heart rate monitor throughout each session (Polar FS2C, Polar Electro Inc., Bethpage, NY) to ensure their heart rate did not exceed light intensity (< 40% heart rate reserve). In addition, RPE (< 12) [20], and the "talk test" [23] were used. In per protocol analyses, the average attendance at the stretching and breathing group was 98.8%. In intention-to-treat analyses, the average attendance was 93.4%.

Maternal Measurements

The previously validated and reliable Modifiable Physical Activity Questionnaire (MPAQ) [24–26] was used at enrollment and after delivery to confirm compliance based on group assignment. This questionnaire also provided participant demographic data, which was also verified with electronic health records. Results from the MPAQ after delivery were used to verify participants' activity levels respective of their assigned group [24, 27]. Importantly, participants in the stretching and breathing group were excluded from the analysis if their activity levels outside of their stretching/breathing routine exceeded the PA recommendations of 150 min/week of moderate activity.

Height was measured in inches to the nearest 0.25 inch using a stadiometer and converted to meters. Using the standard formula for females, BMI was calculated using height, and self-reported pre-pregnancy weight [18]. At 36 weeks gestation, body weight was measured in pounds to the nearest 0.01 pounds using a calibrated scale. Gestational weight gain was calculated by subtracting the participant's pre-pregnancy body weight from the body weight taken at the 36-week visit. Body fat percent at 36 weeks of gestation was estimated with the 5-site skinfold test, as described previously [28, 29].

Infant Measurements

One month after birth, each participant brought her infant in for neurodevelopmental assessment using the Peabody Developmental Motor Scales, 2nd edition (PDMS-2) [30]. The PDMS-2 is a standardized, norm-referenced tool used to assess gross motor skills. For infants up to 12 months of age, the PDMS-2 includes three subtests: Reflexes, Stationary, and Locomotion. The Reflex subtest assesses the child's ability to automatically react to environmental stimuli [30]. The Stationary subtest measures a child's ability to Maintain control of his/her body within the base of support and retain equilibrium [30]. The Locomotion subtest measures a child's

ability to move from one place to another (i.e., rolling over, crawling, walking, running, hopping, jumping) [30]. At one month of age, the PDMS-2 Locomotion subtest primarily assesses early motor behaviors that are precursors to rolling over. While full rolling is not expected at this stage, infants may show initial signs such as head turning or slight trunk rotation when lying in a supine or prone position [30]. A composite percentile derived from the Reflex, Stationary, and Locomotion percentiles, the Gross Motor Quotient (GMQ), and percentile rankings for all subtests and for GMQ, were also calculated [30]. The PDMS-2 was performed and percentiled by an experienced pediatric physical therapist blinded to group classification and percentiled according to the standardized protocol.

Data Analysis

Per protocol and intention-to-treat analyses were conducted to assess the effect of concurrent exercise during pregnancy on 1-month neuromotor skills. To identify possible differences based on maternal pre-pregnancy BMI, intervention and stretching and breathing participants were grouped according to normal-weight (pre-pregnancy BMI from 18.5 to 24.9 kg/m^2) or overweight/obese women (pre-pregnancy BMI ≥ 25.0 kg/m^2).

Maternal and infant descriptive measures were summarized as means ± standard deviations (SD) and compared using either independent t-tests or Pearson chi-square to determine between-group differences.

Pearson correlations were used to determine relationships between maternal and infant PDMS-2 subtest percentiles. To determine between-groups differences on neuromotor percentiles, independent t-tests (in the unadjusted model) and ANCOVA analyses were performed while controlling for potential cofounders after a correlation analysis was made (adjusted model): maternal age, gravida, parity, maternal educational level and body weight gain. The effect sizes were expressed as partial eta-squared (η2p), which describes the ratio of variance explained in the dependent variable by an independent variable as well other controlled variables in the adjusted model. Based on previous recommendations, η2p values were categorized as either small (> 0.01), medium (≥ 0.06), or large (≥ 0.14) [31].

All statistical analyses were performed using the Statistical Package for Social Sciences (SPSS version 19.0, Chicago, 2009).

Results

Sample Description

A flow chart of the study participants is shown in Fig. 1. From the 179 randomized into concurrent exercise ($N = 71$) or stretching and breathing participants ($N = 108$), a total of 94 pregnant women and their infants were included in the present analysis. Forty-three normal-weight women were divided into exercise ($n = 20$; age = 30.9 ± 4.0 years, BMI = 22.8 ± 1.3 kg/m^2) or stretching and breathing group ($n = 23$; age = 30.1 ± 4.0 years, BMI = 21.9 ± 1.6 kg/m^2), and 51 overweight/obese women into exercise ($n = 22$; age = 31.3 ± 4.5 years, BMI = 28.9 ± 3.0 kg/m^2) and stretching and breathing group ($n = 29$; age = 29.8 ± 4.4 years, BMI = 30.8 ± 4.8 kg/m^2).

Sociodemographic characteristics of the study participants are shown in Table 1. Baseline differences between groups were observed in pre-pregnancy BMI and maternal educational level. Consequently, women were stratified by pre-pregnancy BMI groups, and maternal educational level was included in ANCOVA models to avoid the risk of bias as a possible cofounder. No statistically significant between-group differences were found for any of the other variables at baseline ($p \geq 0.05$).

The effects of concurrent exercise training on infant neuromotor skills outcomes are shown in Table 2.

Effect of Supervised-concurrent Exercise Training During Pregnancy on Infant PDMS-2 Reflex Percentile

Only in the unadjusted model, per protocol and intention-to-treat analyses showed that infants from the exercise group and whose mothers were normal-weight, presented better PDMS-2 Reflex percentiles than infants from stretching and breathing participants ($p = 0.044$, $\eta_p^2 = 0.08$; $p = 0.033$, $\eta_p^2 = 0.08$, respectively). No statistically significant differences between groups were found in infants whose mothers were overweight/obese ($p \geq 0.05$).

Effect of Supervised-concurrent Exercise Training During Pregnancy on Infant PDMS-2 Stationary Percentile

Only in the adjusted model, per protocol and intention to treat analyses showed that infants from the exercise group and whose mothers were normal-weight presented better PDMS-2 Stationary percentiles than infants from stretching and breathing participants ($p = 0.022$, $\eta_p^2 = 0.20$; $p = 0.021$, $\eta_p^2 = 0.19$, respectively). No statistically significant differences between groups were found in infants whose mothers were overweight/obese ($p \geq 0.05$).

Effect of Supervised-concurrent Exercise Training During Pregnancy on Infant PDMS-2 Locomotion Percentile

In both the unadjusted and the adjusted model (close to signification), per protocol analyses showed that infants from the exercise group and whose mothers were normal-weight, presented better PDMS-2 Locomotion percentiles than infants from stretching and breathing participants ($p = 0.043$, $\eta_p^2 = 0.08$; $p = 0.054$, $\eta_p^2 = 0.15$, respectively). Intention-to-treat analyses indicated,

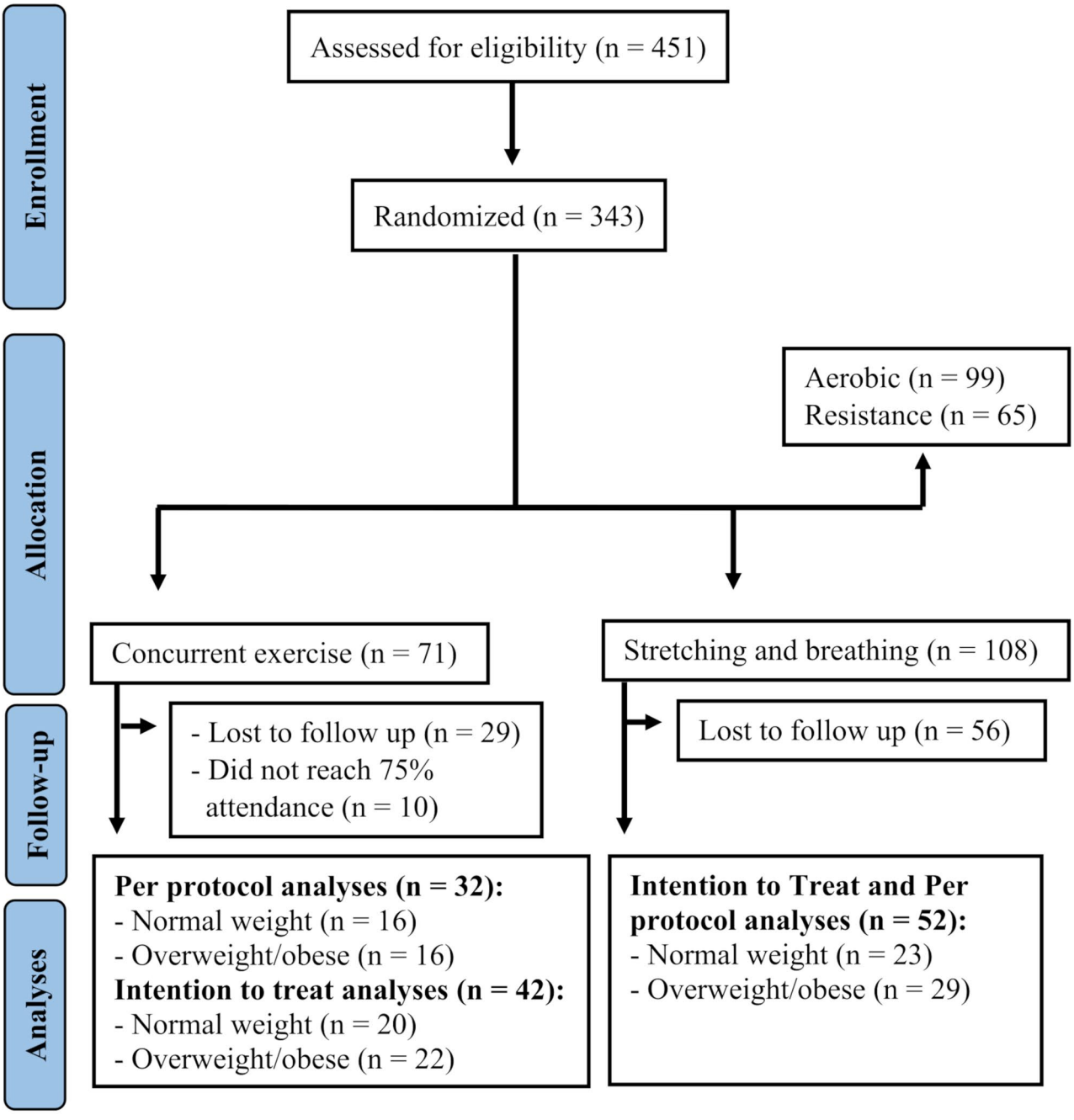

Fig. 1 Flow chart of participants

in both the unadjusted and the adjusted model, that infants from the exercise group and whose mothers were normal-weight presented better PDMS-2 Locomotion percentiles than infants from stretching and breathing participants ($p = 0.024$, $\eta_p^2 = 0.09$; $p = 0.032$, $\eta_p^2 = 0.17$, respectively). No statistically significant differences between groups were found in infants whose mothers were overweight/obese ($p \geq 0.05$).

Effect of a Supervised-concurrent Exercise Training Program During Pregnancy on Infant PDMS-2 Gross Motor Quotient

Only in the adjusted model, per protocol and intention-to-treat analyses showed that infants from the exercise group and whose mothers were normal-weight presented better PDMS-2 GMQ percentiles than infants from stretching and breathing participants ($p = 0.022$, $\eta_p^2 = 0.20$; $p = 0.020$, $\eta_p^2 = 0.19$, respectively). No statistically significant differences between groups were

Table 1 Maternal demographics and pregnancy outcomes according to group assignment

	Per protocol analysis (compliance ≥ 75%)						**Intention to treat analysis**					
	Normal weight			**Overweight/obese**			**Normal weight**			**Overweight/obese**		
	CE (*n* = 16)	**SB (*n* = 23)**	***p*-value**	**CE (*n* = 16)**	**SB (*n* = 29)**	***p*-value**	**CE (*n* = 20)**	**SB (*n* = 23)**	***p*-value**	**CE (*n* = 22)**	**SB (*n* = 29)**	***p*-value**
Demographics												
Maternal age (years)	30.3 ± 4.1	30.1 ± 4.0	0.829	30.9 ± 4.9	29.8 ± 4.4	0.440	30.9 ± 4.0	30.1 ± 4.0	0.535	31.3 ± 4.5	29.8 ± 4.4	0.256
Gravida (n)	1.7 ± 0.8	1.7 ± 1.0	0.979	1.8 ± 0.9	2.0 ± 1.0	0.538	1.8 ± 1.0	1.7 ± 1.0	0.859	1.9 ± 0.8	2.0 ± 1.0	0.607
Parity (n)	0.4 ± 0.6	0.7 ± 0.9	0.307	0.4 ± 0.6	0.9 ± 0.9	0.106	0.4 ± 0.7	0.7 ± 0.9	0.324	0.6 ± 0.7	0.9 ± 0.9	0.178
Educational level (n (%))			0.602			**0.019**			0.793			**0.021**
High School diploma	1 (7.1)	2 (10.5)		2 (13.3)	2 (8.7)		1 (7.1)	2 (10.5)		2 (10.5)	2 (8.7)	
Associates degree	0 (0)	0 (0)		0 (0)	2 (8.7)		0 (0)	0 (0)		0 (0)	2 (8.7)	
Bachelors degree	4 (33.3)	9 (47.4)		2 (13.3)	10 (43.5)		5 (35.7)	9 (47.4)		3 (15.8)	10 (43.5)	
Masters degree	6 (50.0)	5 (26.3)		6 (40.0)	9 (39.1)		6 (42.9)	5 (26.3)		8 (42.1)	9 (39.1)	
Doctorate degree	1 (8.3)	3 (15.8)		5 (33.3)	0 (0)		2 (14.3)	3 (15.8)		6 (31.6)	0 (0)	
Pre-pregnancy BMI (kg*m.^{2})	22.9 ± 1.4	21.9 ± 1.6	0.050	28.2 ± 2.5	30.8 ± 4.8	**0.049**	22.8 ± 1.3	21.9 ± 1.6	**0.041**	28.9 ± 3.0	30.8 ± 4.8	0.10
Pregnancy outcomes												
Gestational Weight gain (kg)	13.8 ± 4.9	12.9 ± 4.9	0.604	13.7 ± 7.1	11.8 ± 5.7	0.343	14.5 ± 4.9	12.9 ± 4.9	0.292	14.3 ± 6.6	11.8 ± 5.7	0.156
Body fat 36 week (%)	33.8 ± 3.8	33.3 ± 2.9	0.699	35.7 ± 3.4	36.8 ± 2.8	0.317	34.0 ± 3.4	33.3 ± 2.9	0.548	36.1 ± 3.6	36.8 ± 2.8	0.522
Physical activity (METs/week)	694.2 ± 195.3	385.3 ± 103.5	**< 0.001**	645.6 ± 244.5	289.5 ± 157.8	**< 0.001**	649.5 ± 199.0	385.3 ± 103.5	**< 0.001**	583.4 ± 250.6	289.5 ± 157.8	**< 0.001**
Gestational age (weeks)	39.2 ± 1.1	39.0 ± 1.6	0.588	39.9 ± 1.0	39.1 ± 1.7	0.132	39.4 ± 1.2	39.0 ± 1.6	0.285	39.6 ± 1.3	39.1 ± 1.7	0.296

Notes: Student's t-tests and Pearson Chi-Square tests used to assess between-group differences in continuous and categorical data; Data presented as mean ± standard deviation as well as n (%). Abbreviations: BMI, body mass index; CE, concurrent exercise; SB, stretching and breathing

found in infants whose mothers were overweight/obese ($p \geq 0.05$).

Discussion

Main Findings

The purpose of this post-hoc analysis was to examine the effect of a supervised-concurrent exercise training program from the 16th gestational week until delivery on 1-month infant neuromotor skills, based on maternal pre-pregnancy BMI. The main findings of the present study suggest that infants from the exercise group and whose mothers were normal-weight had better PDMS-2 Stationary, Locomotion, and GMQ percentiles than infants from stretching and breathing participants. Importantly, these differences were accompanied by large effect sizes, highlighting the potential clinical relevance of the intervention beyond statistical significance. This reinforces the practical importance of the results and supports the interpretation that supervised-concurrent exercise during pregnancy may have a meaningful impact on infant neuromotor skills. Interestingly, no differences between groups were found in infants whose mothers were overweight or obese.

Effect of Supervised-concurrent Exercise Training During Pregnancy on Infant PDMS-2 Reflex Percentile

The mixed results found regarding the PDMS-2 Reflex percentile in the unadjusted and adjusted analyses could be explained since, PDMS-2 Reflex percentiles in the normal-weight group had significantly high variability, and only one item in this subtest is expected to be exhibited at this age. To the best of the investigative team's knowledge, only one previous study to date has assessed the effect of exercise during pregnancy on infant neuromotor skills with the PDMS-2, whose reported results suggested improved one-month neuromotor skills associated with aerobic exercise exposure in utero [16]. Also, the control group had an intervention with a programme of stretching and breathing exercises three days a week. This may have influenced why differences were not greater between the two study groups, as there is some evidence showing that adequate levels of flexibility during pregnancy could induce maternal and neonatal benefits (e.g., reduced risk of requiring exogenous oxytocin administration and cesarean delivery [32], and improved baby's acid-base balance after birth, with more alkaline arterial pH, higher arterial oxygen and lower arterial partial pressure of carbon dioxide in umbilical cord blood [33]).

Therefore, further studies are needed to assess whether PDMS-2 Reflex percentiles of infants are influenced by exercise during pregnancy. In addition, it is suggested that future research examine results when stratified by sex of the child, prenatal exercise mode (e.g., aerobic, resistance or concurrent exercise training), and maternal BMI.

Effect of Supervised-concurrent Exercise Training During Pregnancy on Infant PDMS-2 Stationary Percentile

In the present study, PDMS-2 Stationary percentiles from the infants whose mothers were normal-weight and exercised during pregnancy were higher than percentiles achieved by infants from stretching and breathing participants.

Exercise during pregnancy and a healthy pre-pregnancy BMI contribute to an improved metabolic intrauterine environment [2], reduce the risk of macrosomia (birth weight > 4000 g) [5, 34], and consequently, the risk of developing obesity later in life [35] and deficits in cognition and postural control, among other alterations, in children, adolescents and adults [35]. However, there is a lack of research regarding the potential mechanisms induced by exercise during pregnancy, as well as maternal pre-pregnancy BMI, on offspring motor skill development. Further studies are warranted to confirm or contrast the present findings and the mechanisms behind prenatal exercise and motor skills in childhood.

Effect of Supervised-concurrent Exercise Training During Pregnancy on Infant PDMS-2 Locomotion Percentile

Our findings suggest that infants from the exercise group whose mothers were normal-weight had better locomotion percentiles than stretching and breathing participants, which concurs with McMillan et al. [16] results, as they concluded that infants whose mothers performed aerobic exercise during pregnancy achieved higher PDMS-2 Locomotion percentiles than infants of stretching and breathing participants. Other authors [15, 36] have reported improved gross motor skills in infants whose mothers have performed aerobic exercise during pregnancy compared with stretching and breathing participants, however, these findings were limited to overall gross motor skills.

To our knowledge, there is no study to date that has assessed the effect of a concurrent exercise intervention during pregnancy on PDMS-2 Locomotion percentiles of the offspring.

Effect of a Supervised-concurrent Exercise Training Program During Pregnancy on Infant PDMS-2 Gross Motor Quotient

The positive results found in our study regarding the PDMS-2 GMQ of infants exposed to concurrent exercise in utero and whose mothers were normal-weight compared with stretching and breathing participants aligned with other studies that found improved neuromotor skills in 5-weeks [36] among 1 year-old children [15] whose mothers performed aerobic exercise during pregnancy.

Table 2 Effect of concurrent exercise during pregnancy on neonatal neurodevelopment

Outcomes	Per protocol analysis (compliance ≥ 75%)								Intention to treat analysis							
	Normal weight				Overweight/obese				Normal weight				Overweight/obese			
	CE	SB	*p*	η_p^2	CE	SB	*p*	η_p^2	CE	SB	*p*	η_p^2	CE	SB	*p*	η_p^2
Reflex (%)	65.5 ± 9.7	53.4 ± 26.1			60.4 ± 8.3	64.5 ± 14.7			65.5 ± 10.5	53.4 ± 26.1			61.1 ± 8.1	64.5 ± 14.7		
Unadjusted model			**0.044**	0.08			0.159	-			**0.033**	0.08			0.168	-
Adjusted model			0.121	-			0.629	-			0.130	-			0.575	-
			0.244	-												
Stationary (%)	42.9 ± 13.1	38.9 ± 10.3			39.56 ± 9.5	42.6 ± 12.5			43.7 ± 14.0	38.9 ± 10.3			42.4 ± 12.2	42.6 ± 12.5		
Unadjusted model			0.149	-			0.202	-			0.103	-			0.475	-
Adjusted model			**0.022**	0.20			0.974	-			**0.021**	*0.19*			0.704	-
Locomotion (%)	54.8 ± 9.2	49.0 ± 10.8			51.6 ± 10.2	52.7 ± 8.8			55.2 ± 8.7	49.0 ± 10.8			53.0 ± 9.5	52.7 ± 8.8		
Unadjusted model			**0.043**	0.08			0.357	-			**0.024**	0.09			0.459	-
Adjusted model			0.054	0.15			0.687	-			**0.032**	*0.17*			0.944	-
GMQ (%)	55.4 ± 11.1	51.7 ± 7.4			50.7 ± 9.1	55.3 ± 9.2			55.7 ± 10.8	51.7 ± 7.4			52.8 ± 10.3	55.3 ± 9.2		
Unadjusted model			0.105	-			0.116	-			0.080	-			0.187	-
Adjusted model			**0.022**	0.20			0.058	-			**0.020**	*0.19*			0.744	-

Between-group differences were assessed using independent t-tests for the unadjusted model and analysis of the covariance for the adjusted model. Model-specific covariates included: maternal age, gravida, parity, maternal educational level, body weight gain. Effect size was determined by partial eta squared ($0.01 \leq \eta_p^2 < 0.6$ = small effect, $0.6 \leq \eta_p^2 < 0.14$ = medium effect, $\eta_p^2 \geq 0.14$ = large effect). Abbreviations: CE, concurrent exercise; GMQ, gross motor quotient; SB, stretching and breathing

However, Ellingsen et al. [37] found that 7 years-old children neurodevelopment was not affected by exercise during pregnancy. This result could differ with us because the tool used to assess children's neurodevelopment was a self-reported questionnaire carried out by the parents of the children.

This positive finding could be explained by the improvement of circulation and umbilical cord flow by maternal exercise during pregnancy, contributing to an increased supply of nutrients, oxygen and insulin-like growth factor-1 to the fetal nervous system [16, 38]. Also, it has been published that, mothers who engaged in concurrent exercise during pregnancy exhibited higher levels of mitochondrial DNA content compared with controls, which were associated with greater intelligence in children from 8 to 15 years [39] and could be mediated through an improvement of mitochondrial bioenergetics and antioxidative capacity [40]. Additionally, aerobic exercise during pregnancy has been associated with increased levels of brain-derived neurotrophic factor, an important molecule for neurogenesis and brain development [17] to the fetus.

There are several hypotheses that could explain why we did not find differences between groups in any of the variables assessed regarding infants whose mothers were overweight/obese. From a statistical point of view, the small sample size when subgroups were created makes plausible a lack of statistical power. Additionally, Fernandes et al. [41] observed that overweight/obese children 3–5 years showed worse locomotor skills and GMQ than normal-weight children [41]. The lack of improvement in GMQ with our exercise protocol may suggest that a higher dose, or different type of exercise, might be needed to elicit detectable changes in offspring of overweight/obese women. Hellenes et al. [42] and Leao et al. [43] did assess the effect of a concurrent exercise intervention during pregnancy on gross motor skills in children, both with no statistically significant differences between groups. However, their results were not adjusted or stratified by pre-pregnancy BMI, which makes the current study, to be the first one to our knowledge, to assess the effect of a concurrent exercise intervention during pregnancy on infant neuromotor skills, dependent on pre-pregnancy maternal BMI.

Maternal obesity during pregnancy is associated with poor outcomes in both mothers and their offspring [44, 45]. Mothers with pre-pregnancy obesity are more likely to deliver infants with health concerns (including neurodevelopmental disorders) throughout their life [44]. However, current knowledge regarding the underlying mechanisms of fetal programming remains limited.

A plausible hypothesis regarding our lack of effects of concurrent exercise during pregnancy on infants' neuromotor skills when their mothers were overweight/obese

could be the potential for neuroinflammation mediated by maternal obesity. Although pregnancy itself is characterized by an altered inflammatory profile compared to the non-pregnant state, animal studies have shown that the balance between pro- and anti-inflammatory cytokines is essential for embryonic implantation, formation of the placenta, and continuation of a healthy pregnancy [46–48]. In that sense, maternal obesity is associated with an impaired immune cell profile that leads to activation of a chronic metabolic pro-inflammatory state [49–51], recognized as an early life factor for offspring health [52, 53] that alters placental function (e.g., higher cytokine secretion and transport, poor nutrient transport) [54].

Increased levels of cytokines in the developing fetus affect brain derived neurotrophic factor expression [44], and reduces fractalkine levels [55], which play a crucial role in the homeostasis of the microglial function [56]. In animal models, microglial cells influence neuronal death and the maturation of synaptic circuits at early postnatal stages, controlling brain development of infants [55, 57, 58], and regulating oligodendrocytes generation and myelination throughout life [59]. In humans, Acosta et al. [60] concluded that concurrent exercise during pregnancy seems to have a protective role in attenuating the maternal-fetal proinflammatory status (assessed in maternal serum and umbilical cord serum) associated with excessive gestational weight gain during pregnancy. Additionally, concurrent exercise during pregnancy has been associated with a less pro-inflammatory profile in human milk, especially in colostrum, also by increasing milk fractalkine concentrations [61]. For that reason, concurrent exercise interventions during pregnancy are advisable to reduce the exacerbated pro-inflammatory state in pregnancy and the consequences associated with such. However, overweight/obese women have an altered inflammatory state due to pregnancy and their high levels of adipose tissue, a major source of cytokines [62]. Therefore, it could be advisable to start exercise interventions prior to pregnancy to explore their benefits in offspring neuromotor skills. A review from Sauder et al. [63] suggested that pre-conception interventions to reduce intergenerational obesity seem to be more effective than interventions during pregnancy for reducing maternal weight. However, the majority of pre-conception exercise studies have been designed to improve fertility, and the motivation to lose weight of this population limits generalizability. Further, most results were based on end-of-intervention assessments, which may not have persisted until end of pregnancy [63]. However, pre-conception lifestyle interventions have shown positive benefits on body composition, glucose homeostasis, and insulin resistance [63], which have been associated in pregnancy with a poorer microbiota [64]. Along those lines, both insulin resistance and altered microbiota have been linked to obesity, inflammation, and neurodevelopmental disorders [65, 66]. However, there is a lack of evidence suggesting that lifestyle pre-conception interventions benefit offspring. Further studies of pre-conception (and during pregnancy) exercise interventions with overweight/obese mothers are required to explore their potential benefits on offspring neuromotor skills.

Limitations and Strengths

The current study does have limitations. First, we did not adjust the models by breastfeeding, which could influence the neurodevelopment of infants. We could not do this because, at one month, all our participants reported that they were breastfeeding. Second, we did not analyze maternal diet during pregnancy. Third, at one month, the sample was reduced as we lost to follow-up data from 29 infants in the intervention group, and 56 infants in the stretching and breathing group, which represents the 40.8% and the 51.9% of each group, respectively. Four, type of birth (euthocic, instrumental or caesarean) was not collected, which could have partially influenced the present results, as cesarean delivery has been associated with poorer neurodevelopmental outcomes compared to children born through euthocic delivery [67]. Lastly, this study did not perform analyses based on sex of the child due to a small sample size, which makes impossible to determine if the influence of concurrent exercise during pregnancy on infant neuromotor skills at one month is sex-specific.

The main strength of this clinical trial is that it is the first study to our knowledge that has analyzed the effect of supervised-concurrent exercise training during pregnancy on neuromotor skills in childhood. It is also the first time that the influence of an exercise intervention on infant neuromotor skills was assessed by maternal pre-pregnancy BMI. In addition, the study design is a prospective, randomized controlled trial, which provides the strongest evidence. The results of the current study present are clinically relevant, as the inclusion of concurrent exercise interventions could improve infants' neuromotor skills and prevent future neurodevelopmental diseases.

Interpretation

The main findings of the present study suggest, for the first time, that infants from normal-weight pregnant women who performed a concurrent exercise training program percentiled higher at 1 month of age on PDMS-2 Stationary subtest, Locomotion subtest, and overall GMQ than infants of normal-weight stretching and breathing participants. These findings are of vital importance for public health, as at this age, neurological pathways influencing brain structure and function are developing [9, 10], and concurrent exercise during

pregnancy could lead to physiological improvements [16] that may result in improved infants neuromotor skills.

However, no differences between groups were found in infants whose mothers were overweight/obese, which could need additional inputs, as pre-conception interventions, dietary interventions, greater doses of exercise, or a different type of exercise.

Further studies are compulsory to enlighten the mechanisms behind the effect of concurrent exercise during pregnancy on infant neuromotor skills, and the role of maternal BMI as a mediator of the potential benefits of concurrent exercise on infant neuromotor sills, as well as to know the influence of concurrent exercise during pregnancy throughout childhood at different points of time.

Conclusion

Infants from normal-weight pregnant women who performed a concurrent exercise training program percentiled higher at one month of age on PDMS-2 Stationary subtest, Locomotion subtest, and overall GMQ than infants of normal-weight stretching and breathing participants. However, this exercise intervention had no effect on the descendants from overweight/obese pregnant women who possibly might require additional inputs (e.g., pre-conception interventions, dietary interventions, greater doses of exercise, or a different type of exercise).

Further studies are required to elucidate the specific mechanisms behind the effect of concurrent exercise during pregnancy on infant neuromotor skills, and the role of maternal BMI as a mediator of the potential benefits of concurrent exercise on infant neuromotor sills.

Abbreviations

BMI	Body mass index
GMQ	Gross Motor Quotient
MPAQ	Modifiable Physical Activity Questionnaire
PDMS-2	Peabody Developmental Motor Scales, 2nd edition
RCT	Randomized controlled trial
RPE	Rating of perceived exertion

Supplementary Information

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Supplementary Material 1

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Author Contributions

Sandra Sanchez-Parente led the writing and data analysis of the manuscript. Sandra Sanchez-Parente, Linda E. May, Jose Castro-Piñero and Virginia A Aparicio designed, planned and conducted the study, as well as performed the data analysis. Amy Gross McMillan led the data collection. Sandra Sanchez-Parente, Kara Kern, Alex Claiborne, Breanna Wisseman, Dylan Steen, Brittany Roenker and Ashton Lilley contributed to the data collection and analysis. All authors read and approved the final version.

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Data Availability

The datasets used and/or analysed during the current study are available from the corresponding author on reasonable request.

Declarations

Ethics Approval and Consent to Participate

All protocols were approved on February 13, 2012 by the East Carolina University Institutional Review Board (#12-002425) in accordance with the standards of ethics outlined in the Declaration of Helsinki for experiments involving human participants. All participants signed an informed consent.

Consent for Publication

Not applicable.

Competing Interests

The authors declare that they have no competing interests.

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